



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

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Section: K

Group No: N/A

Project Report On

Project Name [Wholesale Management System]

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Obtained Marks for CO2 and CO3 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO3)		Total =
Requirement fulfillment			Organization of the application		
Validation			Representation and Integration of Database		
Verification			Graphical User Interface		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation of verification processes, and data flow is not considered.	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to ensuring data integrity and appropriate data flow within the system.	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .

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1. Title

SmartWholesale

A Wholesale Management System for enhanced inventory, billing, and customer management.

2. Introduction

SmartWholesale is a desktop-based wholesale management platform designed to streamline the tri-party business model between Admins, Managers, and Customers. Inspired by platforms such as Daraz, Chaldal, and Foodpanda, the system centralizes product management, billing, transaction tracking, and customer reviews into a single, cohesive application.

The platform is built using C# Windows Forms, SQL Server, and ADO.NET, and follows a strict three-layer architecture — UI (Presentation Layer), BLL (Business Logic Layer), and DAL (Data Access Layer) — to ensure clean separation of concerns, maintainability, and scalability. The system is designed to demonstrate all four core pillars of Object-Oriented Programming: Encapsulation, Inheritance, Polymorphism, and Abstraction.

3. Case Study

SmartWholesale supports three distinct user roles, each with a defined set of responsibilities and access rights:

Admin

Rahim is a system Admin who oversees the entire platform. He can add or remove products from the inventory, track total earnings, monitor overall sales figures, and view customer reviews to assess product quality and customer satisfaction. Rahim ensures the platform runs smoothly and that the product catalog is always up to date.

Manager

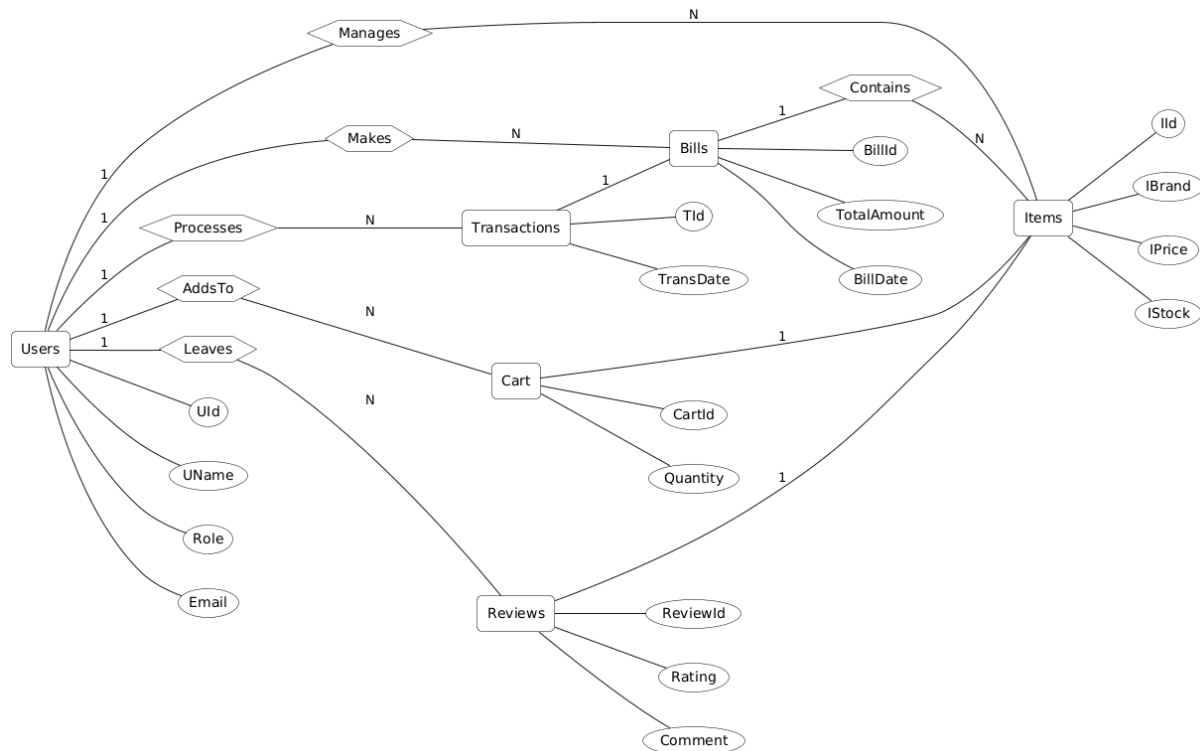
Karim is a Manager who assists with day-to-day inventory operations. He can add or remove products, create discounts on specific items to boost sales, and view customer reviews. Karim acts as the bridge between the administrative oversight and the day-to-day storefront, ensuring pricing is competitive and the stock is well-managed.

Customer

Sadia is a regular customer who browses the SmartWholesale catalog. She can add products to her cart, complete purchases which generate bills and transactions, and leave reviews for items she has bought. Her feedback is visible to both Admins and Managers, helping shape future product and discount decisions.

Together, these three roles form the operational backbone of SmartWholesale — a platform where administration, management, and customer experience are seamlessly integrated.

4. ER Diagram



5. Normalization & Finalization

Following the format of the sample report, here is the normalization process for the core modules:

I. User Management

UNF: UId, UName, UPassword, UPhoneNo, Email, UAddress, Role, JoiningDate, Salary.

1NF: Same as UNF (All values are atomic).

2NF:

1. Users: UId, UName, UPassword, UPhoneNo, Email, UAddress, Role, JoiningDate, Salary.

3NF: Same as 2NF (No transitive dependencies).

II. Inventory (Items)

UNF: IId, IType, IBrand, IModelNo, IPrice, IStockStatus, IMinimumStock, OwnerId.

1NF: Same as UNF.

2NF:

1. Items: IId, IType, IBrand, IModelNo, IPrice, IStockStatus, IMinimumStock, OwnerId (FK).

3NF: Same as 2NF.

III. Billing & Transactions (Normalization through Junction Table)

UNF: BillId, BillDate, TotalAmount, CustomerId, IId, Quantity, UnitPrice, TId, TransactionDate.

1NF: BillId, BillDate, TotalAmount, CustomerId, IId, Quantity, UnitPrice, TId, TransactionDate.

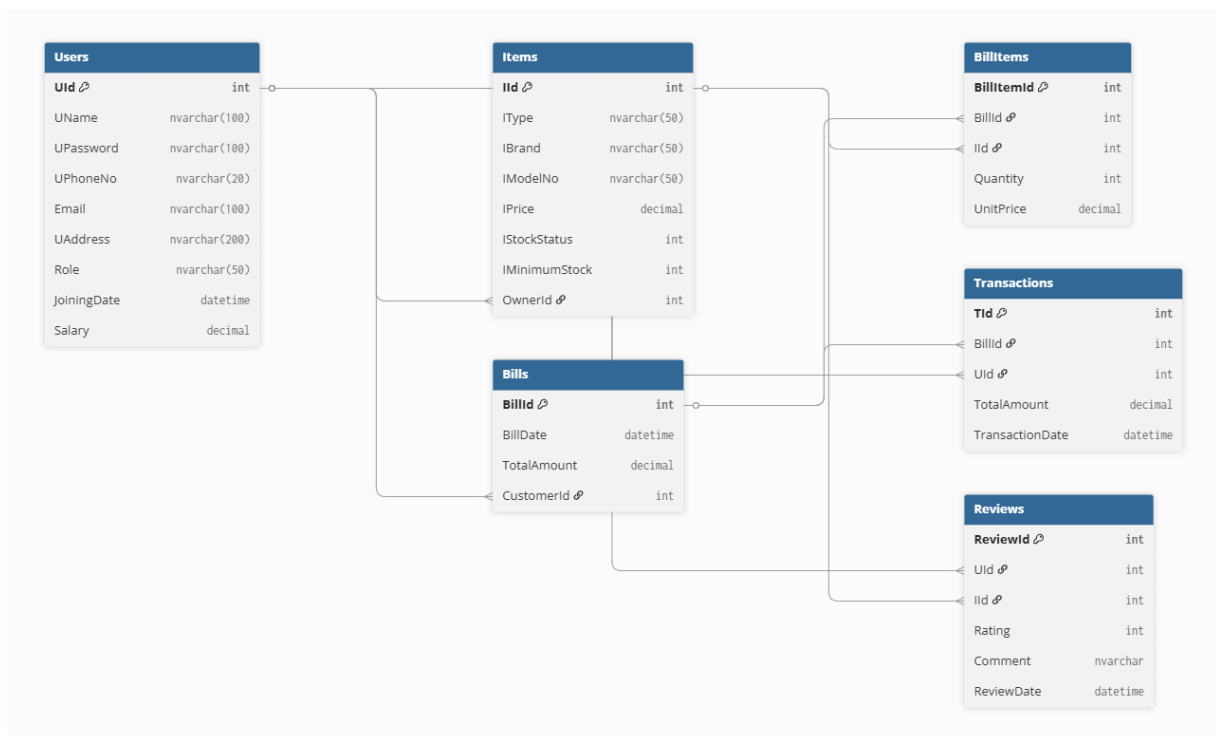
2NF:

1. Bills: BillId, BillDate, TotalAmount, CustomerId (FK).
 2. BillItems: BillItemId, BillId (FK), IId (FK), Quantity, UnitPrice.
 3. Transactions: TId, BillId (FK), UId (FK), TotalAmount, TransactionDate.
- 3NF: Same as 2NF.

Final Tables:

1. Users: [UId (PK), UName, UPassword, UPhoneNo, Email, UAddress, Role, JoiningDate, Salary]
2. Items: [IId (PK), IType, IBrand, IModelNo, IPrice, IStockStatus, IMinimumStock, OwnerId (FK)]
3. Bills: [BillId (PK), BillDate, TotalAmount, CustomerId (FK)]
4. BillItems: [BillItemId (PK), BillId (FK), IId (FK), Quantity, UnitPrice]
5. Transactions: [TId (PK), BillId (FK), UId (FK), TotalAmount, TransactionDate]
6. Reviews: [ReviewId (PK), UId (FK), IId (FK), Rating, Comment, ReviewDate]

6. Database Schema



7. Database Query

DATABASE INITIALIZATION:

- 1 CREATE DATABASE SmartWholesaleDB;
- 2 GO
- 3 USE SmartWholesaleDB;
- 4 GO

Users Table:

```
1 CREATE TABLE Users (  
2   UId INT PRIMARY KEY IDENTITY(1,1),  
3   UName NVARCHAR(100) NOT NULL,  
4   UPassword NVARCHAR(100) NOT NULL,  
5   UPhoneNo NVARCHAR(20),  
6   Email NVARCHAR(100) UNIQUE,  
7   UAddress NVARCHAR(200),  
8   Role NVARCHAR(50) NOT NULL,  
9   JoiningDate DATETIME DEFAULT GETDATE(),  
10  Salary DECIMAL(18, 2) DEFAULT 0);  
12 EXEC sp_columns 'Users';
```

Items Table (Inventory):

```
1 CREATE TABLE Items (  
2   IId INT PRIMARY KEY IDENTITY(1,1),  
3   IType NVARCHAR(50) NOT NULL,  
4   IBrand NVARCHAR(50),  
5   IModelNo NVARCHAR(50),  
6   IPrice DECIMAL(18, 2) NOT NULL,  
7   IStockStatus INT NOT NULL DEFAULT 0,  
8   IMinimumStock INT NOT NULL DEFAULT 5,  
9   OwnerId INT,  
10  FOREIGN KEY (OwnerId) REFERENCES Users(UId));  
12 EXEC sp_columns 'Items';
```

Bills & Transactions:

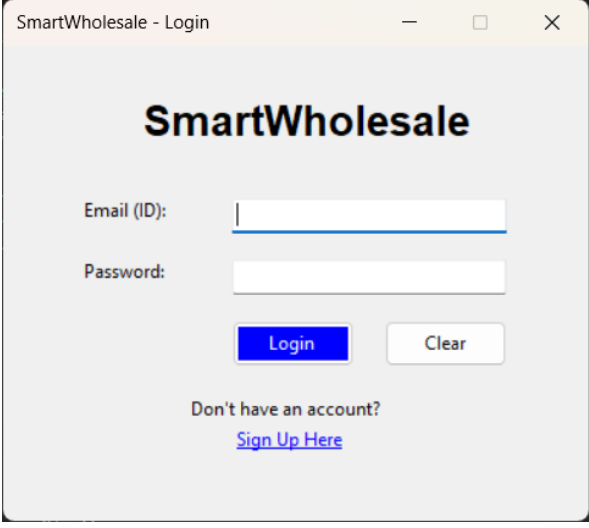
```
1 CREATE TABLE Bills (  
2   BillId INT PRIMARY KEY IDENTITY(1,1),  
3   BillDate DATETIME DEFAULT GETDATE(),  
4   TotalAmount DECIMAL(18, 2) NOT NULL,  
5   CustomerId INT FOREIGN KEY REFERENCES Users(UId);  
6  
7 CREATE TABLE Transactions (  
8   TId INT PRIMARY KEY IDENTITY(1,1),  
9   BillId INT FOREIGN KEY REFERENCES Bills(BillId),  
11  UId INT FOREIGN KEY REFERENCES Users(UId),  
12  TotalAmount DECIMAL(18, 2) NOT NULL,  
13  TransactionDate DATETIME DEFAULT GETDATE());
```

Reviews Table:

```
1 CREATE TABLE Reviews (  
2   ReviewId INT PRIMARY KEY IDENTITY(1,1),  
3   UId INT FOREIGN KEY REFERENCES Users(UId),  
4   IId INT FOREIGN KEY REFERENCES Items(IId),  
5   Rating INT CHECK (Rating BETWEEN 1 AND 5),  
6   Comment NVARCHAR(MAX),  
7   ReviewDate DATETIME DEFAULT GETDATE());
```


8. Transition Table

Login and signup:



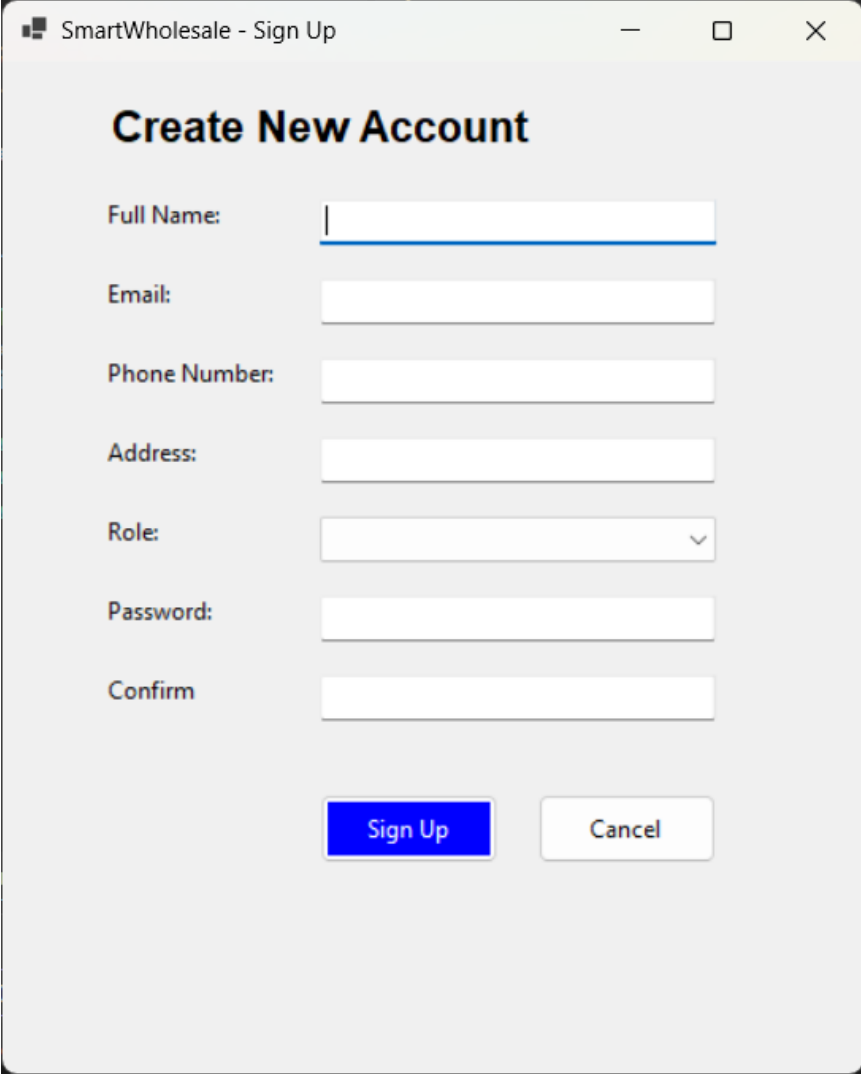
SmartWholesale - Login

SmartWholesale

Email (ID):

Password:

Don't have an account?
[Sign Up Here](#)



SmartWholesale - Sign Up

Create New Account

Full Name:

Email:

Phone Number:

Address:

Role:

Password:

Confirm:

Dashboard (Admin/Shop Owner):

Shop Owner Dashboard - SmartWholesale

Welcome, Shop Owner [Iftekhar Mobin]

!Log Out

My Products

My Earnings

Sales Report

Create Offers

View Reviews

Product & Inventory Management (Admin/Shop Owner):

Product & Inventory Management - SmartWholesale

	Id	Type	Brand	ModelNo	Price	StockStatus	Minimum
▶	5	Car	Porsche	911	20000000.00	1	2
	6	Bike	Suzuki	FI	250000.00	9	3
	7	Smartphone	Xiomi	Redmi Note 13 ...	30000.00	16	5
	8	Laptop	HP	15-fd289TU	77000.00	11	4

Type:Car

Brand:Porsche

Model No:911

Price:20000000.00

Stock:1

Min Stock:2

Add

Update

Delete

Clear

Create Offers (Admin/Shop Owner):

Create Special Offers - SmartWholesale

Create Special Offer / Discount

Item ID:

Discount %:

10

Start Date:

Tuesday , May 5, 2026

End Date:

Tuesday , May 5, 2026

Create Offer

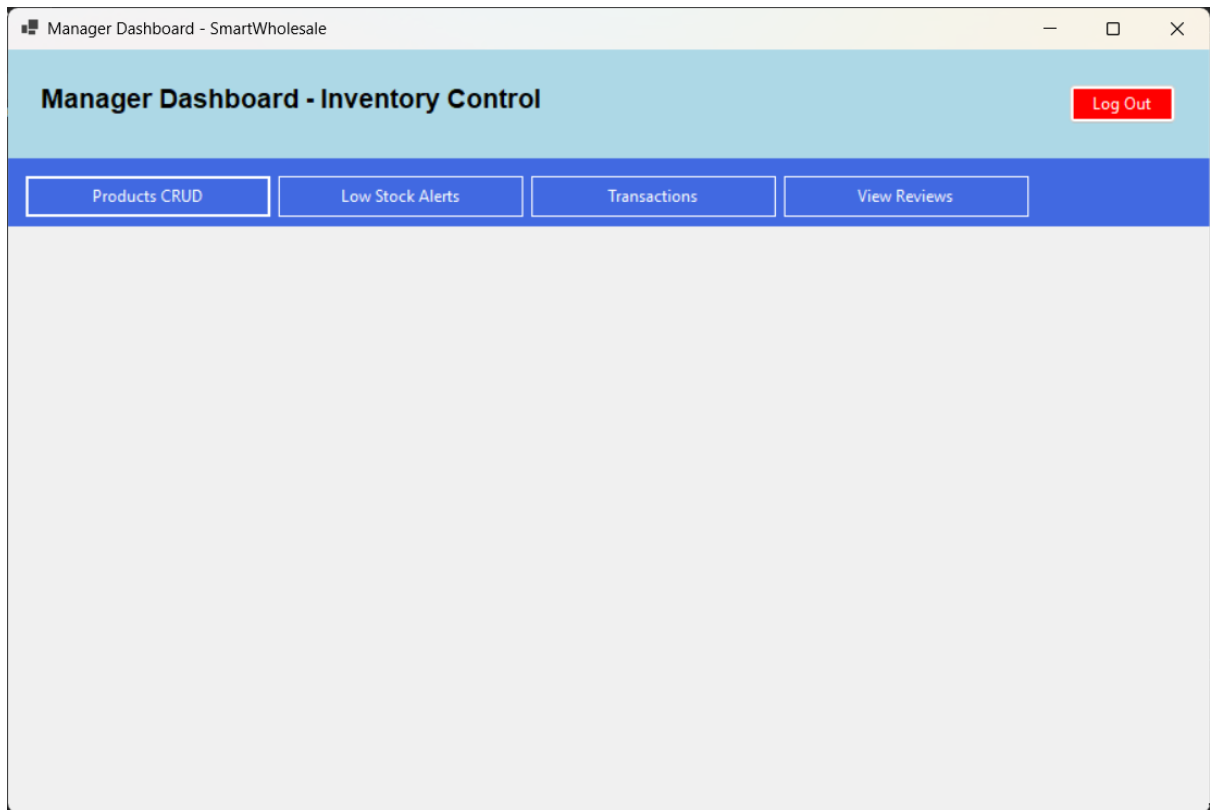
Customer Reviews (Admin/Shop Owner):

Customer Reviews - SmartWholesale

All Customer Ratings and Reviews

	ReviewId	UId	IId	Rating	Comment	ReviewDate	UserName
▶	1	5	8	2	Not worth it	05-May-26 22:16	Siam Tasbir

Dashboard (Manager):



Low Stock Alerts (Manager):

The screenshot shows a web application window titled "Low Stock Alerts - SmartWholesale". The main header is "Items Below Reorder Level" with a red alert message: "Alert: The following items have stock levels below their minimum stock point. Please consider restocking." Below the header is a table with the following data:

	ItemId	ItemType	IBrand	IModelNo	IPrice	IStockStatus	IMinimumStock	Owner
▶	5	Car	Porsche	911	20000000.00	1	2	7

The table is followed by a large grey rectangular area, likely a placeholder for a chart or additional data.

Transactions (Manager):

Transaction Dashboard - SmartWholesale					
Transactions			Search Transaction ID: Search		
	Tid	Billid	UId	TotalAmount	TransactionDate
▶	1	1	5	3500.00	05-May-26 01:48
	2	2	5	20357000.00	05-May-26 02:22
	3	3	5	100000000.00	05-May-26 02:23
	4	4	5	22000.00	05-May-26 02:24
	1002	1002	5	2000.00	05-May-26 09:21
	1003	1003	5	24000.00	05-May-26 10:05
	2002	2002	5	22000.00	05-May-26 22:17

Dashboard (Customer):

Customer Dashboard - SmartWholesale		—	□	×
Welcome, Siam Tasbir		Cart	Log Out	
Browse Products	My Cart	My Orders	My Reviews	

Shopping Cart (Customer):

My Shopping Cart - SmartWholesale

My Shopping Cart

	CartId	UId	IId	Quantity	ItemName	Price
▶	2002	5	2	1	Samsung EVO 2...	2000.00
	2003	5	3	1	GIGABYTE REV...	1500.00
	2004	5	7	1	Xiomi Redmi N...	30000.00

Remove Item

Confirm Purchase

Subtotal: \$33,500.00

Total: \$33,500.00

Reviews (Customer):

Reviews - SmartWholesale

Rate Your Purchase

Select Product: Ryzen 7 5800X

Rating (1-5 Stars): 5

Your Review:

Submit Review

9. Features and Facilities

Admin Features

- Add new products to the inventory catalog.
- Remove existing products from the system.
- View total earnings and overall sales figures through a summarized dashboard.
- View all customer reviews submitted for any product.

Manager Features

- Add new products and update existing product details.
- Remove products from the catalog.
- Create and apply discounts on selected products.
- View customer reviews to monitor product performance.

Customer Features

- Browse the full product catalog with pricing and stock information.
- Add items to a personal cart and adjust quantities.
- Complete purchases, which automatically generate a bill and a linked transaction record.
- Submit reviews (rating + comment) for purchased products.

System-Wide Features

- Role-based access control enforced across all three layers.
- Three-layer architecture (UI → BLL → DAL) ensuring clean separation of concerns.
- ADO.NET-based data access with parameterized queries to prevent SQL injection.
- Real-time stock status updates upon purchase completion.
- Minimum stock threshold alerts to notify managers of low inventory.

10. Conclusion

SmartWholesale successfully demonstrates a production-grade, desktop-based wholesale management system built with C# Windows Forms, SQL Server, and ADO.NET. By adopting a tri-party business model encompassing Admins, Managers, and Customers, the platform mirrors the operational complexity of real-world wholesale and retail platforms.

The system's adherence to a strict three-layer architecture ensures a clean separation of concerns, making the codebase maintainable, scalable, and testable. The implementation of all four OOP pillars — Encapsulation, Inheritance, Polymorphism, and Abstraction — further demonstrates sound software engineering practice.

The normalized database design, coupled with well-defined foreign key relationships and SQL constraints, ensures data integrity and eliminates redundancy. SmartWholesale provides a reliable, intuitive, and extensible foundation for wholesale business management, and stands as a comprehensive demonstration of the concepts covered in OOP2.